

Question - Minimum Fencing

A farmer wants to build a rectangular field with an area of

$$100 \text{ m}^2.$$

The length of the field is x meters, and the width is y meters.

Using Lagrange multipliers, find the dimensions x and y that require the minimum amount of fencing.

Step 1: Objective Function

The amount of fencing needed is the **perimeter**:

$$P = 2x + 2y$$

We want to minimize:

$$P(x, y) = 2x + 2y$$

Step 2: Constraint

The area is 100 m^2 :

$$xy = 100$$

or

$$g(x, y) = xy - 100 = 0$$

Step 3: Lagrangian

$$L = 2x + 2y - \lambda(xy - 100)$$

Step 4: Partial Derivatives

$$\frac{\partial L}{\partial x} = 2 - \lambda y = 0$$

$$\frac{\partial L}{\partial y} = 2 - \lambda x = 0$$

$$\frac{\partial L}{\partial \lambda} = xy - 100 = 0$$

So we have:

$$2 = \lambda y$$

$$2 = \lambda x$$

Since both equal 2,

$$\lambda x = \lambda y$$

Therefore,

$$x = y.$$

Step 5: Use the Constraint

Since

$$xy = 100$$

and

$$x = y,$$

we get

$$x^2 = 100.$$

Thus,

$$x = 10$$

and

$$y = 10.$$

We take the positive values because dimensions cannot be negative.

Final Answer

$$x = 10 \text{ m}, \quad y = 10 \text{ m}$$

So the farmer should build a **10 m × 10 m square**.

The minimum fencing required is

$$P = 2(10) + 2(10) = 40 \text{ m}.$$

Therefore, a square gives the minimum fencing for a rectangular field with a fixed area.